

LiDAR-IMU Fusion System with Adaptive Scanning for High-Resolution Deformation Monitoring of Underground Infrastructures*

Menggang Li, Zhuoqi Li, Kun Hu, Eryi Hu, Chaoquan Tang[†], and Gongbo Zhou[†]

Abstract— A LiDAR-IMU fusion system utilizing adaptive scanning is developed for high-resolution deformation monitoring of underground coal mine infrastructure, such as sealed walls. The system integrates data from a LiDAR scanner and an IMU, employing a penalty function-based scanning strategy to optimize point cloud quality. Following feature extraction and state estimation, a 3D point cloud model of the sealed wall is constructed. Deformation monitoring is achieved through point cloud segmentation, registration, and error analysis across multiple time intervals. A methodology for optimizing equipment placement on walls of varying dimensions is proposed to efficiently capture deformation details. Two metrics, PATD and PARE, are introduced to evaluate system performance. Calibration experiments using standardized boards and blocks are designed to determine optimal monitoring parameters, including distance, height, and sampling frequency. Simulated deformation experiments under real-world conditions validate the system's rationality and accuracy.

I. INTRODUCTION

In recent years, the westward shift of China's energy strategy has accelerated the exploitation of coal resources in western regions, posing significant challenges to mine water hazard prevention [1]. To ensure safety after mining cessation, waterproof sealed walls are typically constructed in areas where mining operations have ceased. Deformation in these underground sealed walls primarily arises from factors such as ground pressure, geological shifts, and mining-induced stresses, leading to structural issues including cracks, misalignment, and tilting [2].

High-precision LiDAR can theoretically measure deformation accurately. However, underground mines face three limits: heavy post-processing, high costs, and no

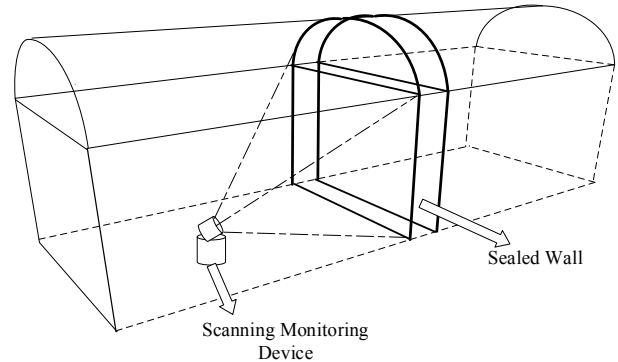


Fig. 1. Schematic diagram of sealed wall monitoring.

real-time monitoring due to discontinuous data collection. They need to process lots of point cloud data through steps like stitching and denoising. For example, Cyclone software mentioned in [3] requires much computation for data registration. Also, lack of an accuracy model in [4] makes post-processing more complex. Processing large data in mines is time consuming [5]. High-end scanners like Leica HDS C10, Trimble GX 200, and FARO X330 are costly to buy and maintain [6]. Compared to traditional devices like total stations, they have higher overall costs. Although they can get single-scan data quickly, mining deformation needs multi-period data analysis. The time-phased acquisition causes delays, not real-time feedback [7]. The "real-time display system" in [8] relies on sensor arrays, not continuous scanning. Our system overcomes these by using edge computing for faster point cloud processing and cost-effective hardware, keeping subcentimeter accuracy.

In light of these challenges, this paper presents an adaptive LiDAR-IMU fusion system for high-resolution dynamic monitoring of deformation and damage in underground coal mine infrastructure. The main contributions are:

- A dynamic high-resolution integrated above and below-ground deformation monitoring system for coal mine-sealed walls has been developed.
- A LiDAR scanning method quickly gets high resolution 3D data of the sealed wall surface, accurately calculating deformation and trends.
- A penalty-function-based adaptive dynamic point cloud scanning strategy adjusts parameters based on sealed wall geometry and scanning conditions, sig-

*Research supported by National Natural Science Foundation of China (grant number: 52304183, 52274159), the China Postdoctoral Science Foundation (2022M723392), The Jiangsu Provincial Key Research and Development (R&D) Plan Projects (BE2023008-4). The Project Funds of the Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD).

Menggang Li is with School of Mechatronic Engineering, China University of Mining and Technology, Jiangsu Collaborative Innovation Center of Intelligent Mining Equipment, China University of Mining and Technology, and National Key Laboratory of Intelligent Mining Equipment Technology, Xuzhou, 221116, China. E-mail: sallylmg@cumt.edu.cn.

Zhuoqi Li, Kun Hu, Chaoquan Tang, and Gongbo Zhou are with School of Mechatronic Engineering, China University of Mining and Technology, Xuzhou, 221116, China. E-mail: {ts24050030a311d, ts22050017a31, tangchaoquan, gbzhou}@cumt.edu.cn.

Eryi Hu is with Information Institute, Ministry of Emergency Management of the People's Republic of China, Beijing 100029, China. E-mail: horyhu@126.com.

Chaoquan Tang and Gongbo Zhou are the common corresponding authors.



2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)
October 26, 2025

CERTIFICATE OF CONFERENCE ATTENDANCE

To Whom It May Concern,

This letter serves as official confirmation that Kun Hu attended the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025). The conference took place from October 19 to October 25, 2025, in Hangzhou, China.

Kun Hu actively participated in various sessions and workshops during the conference, gaining valuable insights into the latest advancements in intelligent robotics and systems. The involvement in the event underscores a commitment to professional development and contribution to the field.

Should you require any further information or verification, please do not hesitate to contact us.

Sincerely,

Wang hesheng



Hesheng Wang

General Chair | IROS 2025

Email: wanghesheng@sjtu.edu.cn

Contact Phone: +86-2134207252

Address: Room 2-509, SEIEE Buildings, Shanghai Jiao Tong University,
Minhang District, Shanghai 200240, China

Conferences > 2025 IEEE/RSJ International C... ?

LiDAR-IMU Fusion System with Adaptive Scanning for High-Resolution Deformation Monitoring of Underground Infrastructures

Publisher: IEEE

Cite ThisPDF

Menggang Li; Zhuoqi Li; Kun Hu; Eryi Hu; Chaoquan Tang; Gongbo Zhou

All Authors

12

Full

Text Views



Back to Results | Next >



Abstract

Document Sections

I. Introduction

II. System Integration

III. Establishment and Matching of Point Cloud Models for

Abstract:

A LiDAR-IMU fusion system utilizing adaptive scanning is developed for high-resolution deformation monitoring of underground coal mine infrastructure, such as sealed walls. The system integrates data from a LiDAR scanner and an IMU, employing a penalty function-based scanning strategy to optimize point cloud quality. Following feature extraction and state estimation, a 3D point cloud model of the sealed wall is constructed. Deformation monitoring is achieved through point cloud segmentation, registration, and error analysis across multiple time intervals. A methodology for optimizing equipment placement on walls of varying dimensions is proposed to efficiently capture deformation details. Two metrics, PATD and

More Like This

Edge-Oriented Point Cloud Compression by Moving Object Detection for Realtime Smart Monitoring

2024 IEEE 21st Consumer Communications & Networking Conference (CCNC)

Published: 2024

Real-Time Scene-Aware Point Cloud Compression

Feedback